

Practical Astronomy Intensive — Entry Check

Practical Astronomy Intensive

Name _____

Date _____

26 questions 40 minutes

Sat before Day 1. Twenty-six questions on what the camp assumes you already read in the quiz study material — and nothing on what it will teach you. The distractors are all real wrong models, so the marked paper names the cohort's weakest assumption rather than just ranking the students.

*Circle one letter for each multiple-choice question. Write short answers on the lines. Tasks marked **Practical** are done, not written — your assessor will watch.*

- 1 You are setting up in the field and want to mark due east on the ground. Someone says 'easy — just note where the Sun came up this morning.' When is that actually correct?
 - a Every day of the year — the Sun always rises due east and sets due west
 - b Never — the Sun's rising point is random from day to day
 - c Any day, provided you are in the northern hemisphere
 - d Only near the two equinoxes; the rest of the year sunrise drifts north or south of due east
- 2 Without any spacecraft, photograph or modern instrument, which single observation most directly shows that the Earth is a sphere rather than a disc?
 - a The fact that people in the southern hemisphere do not fall off
 - b Earth's shadow on the eclipsed Moon is always a circular arc, from every eclipse and every angle
 - c Nobody knew until Columbus sailed and failed to fall off the edge
 - d Photographs taken from orbit — you genuinely cannot establish it from the ground
- 3 The camp site is given as 23°N, 77°E. Which statement about that pair of numbers is correct?
 - a 23°N is measured along a line running north–south, and 77°E along a line running east–west
 - b Both the parallels of latitude and the meridians of longitude are parallel lines that never meet
 - c 23°N is measured north from the equator, which Earth's spin fixes; 77°E is measured from a prime meridian that people agreed on
 - d Greenwich is the natural zero for longitude in the same way the equator is the natural zero for latitude

- 4 You are outside with no chart and want to tell whether a bright point of light is a planet or a star. What is the most reliable naked-eye clue?
- a Planets appear as small discs to the naked eye, while stars appear as points
 - b Planets are always the brightest objects in the sky, so brightness alone settles it
 - c It shines with a steady light while nearby stars twinkle, and over weeks it drifts against the constellation pattern
 - d Planets are always visible somewhere in the sky on any given night
- 5 Are the seven bright stars that make up the figure of Orion close to one another in space?
- a Yes, and that is why every culture independently drew the same figures
 - b Yes — a constellation is exactly the joined-up figure, so anything in it belongs to it
 - c No — they lie at wildly different distances and only line up as a pattern from our particular viewpoint
 - d Yes — a constellation is a group of stars that formed together and travel together
- 6 What is the ecliptic?
- a The circle on the sky where eclipses are visible from Earth
 - b The Sun's yearly path against the background stars — the line the Moon and planets also stay close to
 - c The daily path the Sun traces across the sky from rising to setting
 - d The plane of the Moon's orbit around the Earth
- 7 What makes the Moon a crescent one week and a gibbous the next?
- a The Earth's shadow falls across the Moon and covers part of it
 - b The Moon genuinely changes shape, thinning and fattening through the month
 - c Clouds and the thickness of the atmosphere hide part of the Moon from us
 - d The Sun always lights exactly half the Moon; the phase is how much of that lit half is turned toward us as the Moon goes round
- 8 A student says 'we cannot schedule Moon work before sunset, the Moon is a night-time object.' Is that right?
- a No — the Moon is above the horizon in daylight about half the time; a first-quarter Moon is already high in the afternoon
 - b No — but only because a full Moon can occasionally be caught just before sunset
 - c Yes — the Moon only rises once the Sun has set
 - d Yes — the Moon is far too faint to be seen while the Sun is up, at any phase

- 9 The face of the Moon we never see from Earth is often called the 'dark side'. What is wrong with that name?
- a It is wrong because there is no fixed far side at all — the Moon turns and we see all of it over a month
 - b Nothing about it is dark — it gets exactly as much sunlight as the near side, and is fully lit at new moon. It is unseen, not unlit
 - c Nothing is wrong; it never receives sunlight, which is precisely why it is called dark
 - d The name is fine — the whole Moon is dark, since it makes no light of its own
- 10 A totally eclipsed Moon, deep inside Earth's umbra, glows a dull copper red rather than going black. Why?
- a It should go black; any red is just the eye tiring or the camera adding colour
 - b The Earth's shadow is itself reddish, and it tints whatever it falls on
 - c The Moon's own iron-rich rock glows red once the sunlight is cut off
 - d Sunlight bent through Earth's atmosphere still reaches it — the red is every sunrise and sunset on Earth at once
- 11 The Moon passes between Earth and Sun every month at new moon, and through Earth's shadow line every month at full moon. So why don't we get a solar eclipse and a lunar eclipse every single month?
- a The Moon's orbit is tilted about 5° to the ecliptic, so most months it passes above or below the line and the shadows miss
 - b We do get one of each every month — most are simply too small or too brief for anyone to notice
 - c Earth's rotation carries us out of the way before the shadow can arrive
 - d The Moon's orbit is a long ellipse, so it is usually much too far away for its shadow to reach
- 12 'It is summer in India because Earth is closer to the Sun in those months.' What single observation kills this explanation outright?
- a Distance is shared by the whole planet, so it would make both hemispheres summer at once — yet India's summer is Australia's winter
 - b Earth's orbit is a perfect circle, so its distance to the Sun never changes at all
 - c Nothing kills it; the distance explanation is essentially correct
 - d In summer the tilt leans the whole Earth toward the Sun, which is a different mechanism
- 13 How does the Sun compare with the stars visible on a clear night?
- a The Sun is by far the largest and hottest object of its type known
 - b The Sun is a star, but a uniquely small and cool one — a runt among stars
 - c The Sun is a fundamentally different kind of object from a star; stars burn, the Sun shines
 - d It is an ordinary star — unremarkable in size and temperature — that looks utterly different only because it is so much nearer

- 14 A student writes 'Proxima Centauri is 4.2 light-years away, so it takes 4.2 years to get there.' Two things are wrong with that sentence. Name them in one line.
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- 15 White sunlight goes into a prism and a band of colours comes out, red through violet. What has the prism done?
- a Separated colours that were already present — each wavelength is bent by a different amount, violet the most
 - b Reflected different colours off its faces at different angles
 - c Broken the white light into new colours that did not exist before it entered
 - d Added colour to the light; the glass tints white light as it passes through
- 16 Two telescope objectives have the same focal length, but one mirror is twice the diameter of the other. What does the extra diameter buy you?
- a Twice the magnification, since a bigger objective magnifies more
 - b Twice the light, since the light gathered scales with the diameter
 - c Four times the light collected — the area scales with the square of the diameter — and a finer resolving limit
 - d A brighter image but a larger one too, since the image scale grows with the objective
- 17 Star A has apparent magnitude 1. Star B has apparent magnitude 5. Which looks brighter in the sky?
- a Star B — 5 is five times brighter than 1
 - b Neither — apparent magnitude describes colour, not brightness
 - c Star B — 5 is a bigger number, so it is the brighter star
 - d Star A — the scale runs backwards, and smaller numbers mean brighter
- 18 The faintest stars a good naked eye can catch from a dark site are about magnitude +6. Vega sits at about magnitude 0. How much more light reaches your eye from Vega?
- a About 12 times more — each magnitude doubles the brightness
 - b About 6 times more, since the difference is 6 magnitudes
 - c About 100 times more — 6 magnitudes is defined as exactly $100\times$
 - d About 250 times more — 5 magnitudes is a factor of 100, and each further magnitude multiplies by about 2.5

- 19 Putting the Sun at the centre instead of the Earth was accepted because it explained, cleanly, things the geocentric model could only patch. Which pair is the strongest evidence?
- a The geocentric model was simply unable to predict where the planets would be, so its forecasts were useless
 - b Telescopes revealed the Earth visibly moving against the background stars
 - c Retrograde loops fall out for free as Earth overtakes an outer planet, and Venus shows a full cycle of phases including a small gibbous
 - d Only the heliocentric model can account for day and night, and for the seasons
- 20 Kepler's second law says a planet sweeps out equal areas in equal times. What does that mean for the planet on the ground?
- a Every planet sweeps out the same area per unit time as every other planet
 - b It covers equal distances along its path in equal times
 - c It moves at a constant speed, which is what keeps the swept areas equal
 - d It moves fastest at perihelion, closest to the Sun, and slowest at aphelion — its speed is not constant round the orbit
- 21 Every textbook diagram of the solar system is a lie about one thing in particular. Say in one line what a to-scale diagram would look like instead, and why no book prints one.
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- 22 Why does Charles Messier's catalogue of 110 objects exist at all?
- a He was a comet hunter, and it is a list of fuzzy things that are NOT comets — kept so he would stop being fooled by them
 - b It is a complete catalogue of every deep-sky object visible from Earth
 - c The objects share a common physical nature, which is why they were grouped together
 - d It was intended from the start as a best-of list of the finest deep-sky objects for amateurs
- 23 Why does the Milky Way appear as a band across the sky rather than a patch, or a spiral, or an even glow?
- a It is a distant spiral galaxy that happens to be edge-on to us
 - b We sit at the centre of the galaxy, and the band is the crowded core surrounding us
 - c It is a band of gas and dust in Earth's upper atmosphere lit by starlight
 - d We sit inside a flat disc, so looking along the disc we see stars piled deep in a line, and looking out of it we see few

- 24** M31, the Andromeda Galaxy, is naked-eye visible from a dark site. What are you actually looking at?
- a A single very bright and very distant star, blurred by the atmosphere
 - b A large cloud of gas within the Milky Way, a few thousand light years off
 - c A separate island of hundreds of billions of stars, about 2.5 million light years away and larger than our own galaxy
 - d A dense cluster of stars orbiting the Milky Way, like a big globular
- 25** On an H-R diagram, about 90% of stars fall along a single diagonal band, the main sequence. What does a star's position along that band tell you?
- a Nothing in particular — the band is just where the common stars happen to land
 - b Its distance from us — the diagram is essentially a distance plot
 - c Its age — stars start at the bottom of the band and climb it as they grow older
 - d Chiefly its mass — the band runs from hot, luminous, massive stars at one end to cool, faint, low-mass ones at the other
- 26** A star of 20 solar masses starts with far more hydrogen fuel than the Sun. Does it therefore stay on the main sequence longer? Answer and justify in one line.
